WCN UNIT - 1

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Introduction to Wireless Communication Systems

This section provides a foundational understanding of wireless communication, beginning with its historical evolution, which was driven by the inherent limitations of early systems. It establishes the core principles of radio transmission and the regulatory frameworks that govern the use of the radio spectrum, a finite global resource.

The Evolution of Mobile Radio Communications

The development of mobile radio communication is a story of incremental innovation, where each technological step addressed the shortcomings of its predecessor. The timeline below highlights the key milestones that led from simple, one-way dispatch systems to the complex, high-capacity cellular networks of today.[1]

- **1934:** The first major mobile radio system was the police radio, a conventional, push-to-talk dispatch system that enabled one-way communication from a central point to mobile units.[1]
- **1935:** Edwin Armstrong demonstrated Frequency Modulation (FM), a significant breakthrough that allowed for higher-fidelity audio transmission compared to Amplitude Modulation (AM), laying the groundwork for improved voice quality in future radio systems.[1]
- **1946:** The first public Mobile Telephone Service was launched. This system was rudimentary, often requiring an operator to manually connect calls, and suffered from very limited capacity.[1]
- **1960s:** The introduction of the Improved Mobile Telephone Service (IMTS) represented a significant upgrade, offering full-duplex capability, which allowed users to talk and listen simultaneously, mimicking a standard telephone call.[1]
- **1968:** Bell Labs introduced the revolutionary concept of a cellular mobile system. This was a paradigm shift from the high-power, large-coverage-area model to a low-power system of smaller, adjacent coverage areas called "cells." This concept is the foundation of all modern mobile networks.[1]
- **1976:** The Bell Mobile phone service was launched but demonstrated the critical flaws of non-cellular approaches. Due to its limited number of channels over a large area, it suffered from severe call blocking, rendering it impractical for widespread public use.[1]
- **1983:** The first commercial cellular system, the Advanced Mobile Phone System (AMPS),

was deployed in the United States. It was an analog system based on Frequency Division Multiple Access (FDMA) and the cellular concept proposed by Bell Labs.[1]
- **1991:** The transition to digital technology began with standards like the U.S. Digital Cellular (USDC) IS-54, which used Time Division Multiple Access (TDMA) to increase capacity over the existing AMPS framework.[1]
- **1993:** The IS-95 standard was introduced, bringing Code Division Multiple Access (CDMA) into the commercial cellular market. CDMA offered further capacity gains and inherent features like improved security and resistance to interference, utilizing advanced modulation schemes like Quadrature Phase-Shift Keying (QPSK) and Binary Phase-Shift Keying (BPSK).[1]

Rationale for Modern Wireless Systems: Overcoming Conventional Limitations

The evolution from conventional mobile radio to modern cellular systems was not arbitrary; it was a direct engineering response to a set of fundamental problems that made early systems inefficient and unscalable for mass-market adoption. The cellular concept, specifically the principle of frequency reuse, was conceived to solve these exact issues.[1]

- **Limited Service Capability:** Conventional mobile telephone systems operated on a high-power, large-coverage-area model. A single powerful transmitter would cover an entire city or region. This approach had two major flaws: first, if a user moved from one coverage zone to another, the call would be dropped; second, the number of active users was strictly limited to the number of channels assigned to that entire zone, creating a severe capacity bottleneck.[1]
- **Poor Service Performance:** Because of the limited number of channels available for a large population of subscribers, the probability of a call being blocked (i.e., the user receiving a busy signal because all channels were occupied) was extremely high, especially during busy hours. This resulted in a frustrating and unreliable user experience.[1]
- **Inefficient Frequency Spectrum Utilization:** This was the most critical failure of conventional systems. A radio channel, once assigned to a user, could not be used by anyone else in the entire coverage area for the duration of the call. This meant that a single conversation would occupy a valuable slice of the radio spectrum across an entire metropolitan area, even though the users were only in one specific location. The spectrum, a finite resource, was therefore used in a highly inefficient manner.[1]

The cellular concept directly addressed the problem of inefficient spectrum utilization. By dividing a large coverage area into smaller cells, each with a low-power transmitter, a set of frequency channels could be used in one cell and then be reused in another cell some distance away without causing interference. This single innovation of **frequency reuse** shattered the capacity limits of conventional systems and made the mobile phone a viable technology for the general public.[1]

Fundamental Concepts of Radio Transmission

At its core, all radio communication can be categorized by the directionality and timing of its transmissions. The method used to achieve two-way communication is a defining characteristic of a wireless system.[1]

- **Classification of Transmission Systems:**
  - **Simplex:** Communication occurs in only one direction. The transmitter sends and the receiver only receives. Examples include broadcast radio, television, and basic paging systems.[1]
  - **Half-Duplex:** Communication is two-way, but not simultaneous. A single radio channel is shared for both transmitting and receiving. When one party transmits, the other must listen. This is often managed by a "push-to-talk" mechanism, as seen in walkie-talkies.[1]
  - **Full-Duplex:** Communication is two-way and simultaneous, allowing for a natural, telephone-like conversation where both parties can talk and listen at the same time. This is the standard for all modern mobile phone systems.[1]
- Duplexing Techniques for Full-Duplex Systems:
  To achieve full-duplex communication, the system must ensure that the transmitted signal does not interfere with the received signal. Two primary techniques are used to accomplish this:
  - **Frequency Division Duplex (FDD):** FDD uses two separate radio frequency channels. One channel, the **Forward Channel** (or downlink), is used for transmissions from the base station to the mobile station. The other channel, the **Reverse Channel** (or uplink), is used for transmissions from the mobile station to the base station. Because the transmissions occur on different frequencies, they can happen simultaneously without interfering with each other.[1]
  - **Time Division Duplex (TDD):** TDD uses a single radio frequency channel but divides access to it into alternating time slots. The base station transmits to the mobile in some time slots, and the mobile transmits to the base station in other time slots. These slots alternate so rapidly that the communication appears simultaneous to the user. TDD is more spectrally flexible than FDD but requires precise timing synchronization between the base station and the mobile unit.[1]

Frequency Regulation for Radio Transmission

The radio frequency spectrum is a limited, natural resource that must be managed globally to prevent interference between different services and users. This management is handled by a hierarchy of international and national bodies.[1]

- **International Telecommunications Union (ITU):** Based in Geneva, the ITU is a specialized agency of the United Nations responsible for the worldwide coordination of all telecommunication activities, including the allocation of radio spectrum and satellite orbits.[1]
- **ITU Regions:** The ITU's Radio Communication Sector (ITU-R) has divided the world into three regulatory regions to manage spectrum allocation:
  - **Region 1:** Europe, Africa, the Middle East, and the former Soviet Union.
  - **Region 2:** North and South America, and Greenland.
  - **Region 3:** The Far East (Asia), Australia, and New Zealand.[1]
- **National and Regional Regulators:** Within these broad regions, national or regional bodies create more specific regulations. Prominent examples include the **Federal Communications Commission (FCC)** in the United States and the **Conference for European Posts & Telecommunications (CEPT)** in Europe.[1] These bodies are responsible for licensing spectrum to service providers and setting technical standards within their jurisdictions.

Pre-Cellular Wireless Technologies

Before the widespread adoption of cellular networks, several other wireless technologies provided limited mobility and messaging services. These systems, while not as capable as cellular, were important technological stepping stones, introducing the public to the concepts of personal wireless communication and paving the way for future advancements.

Paging Systems [FIG]

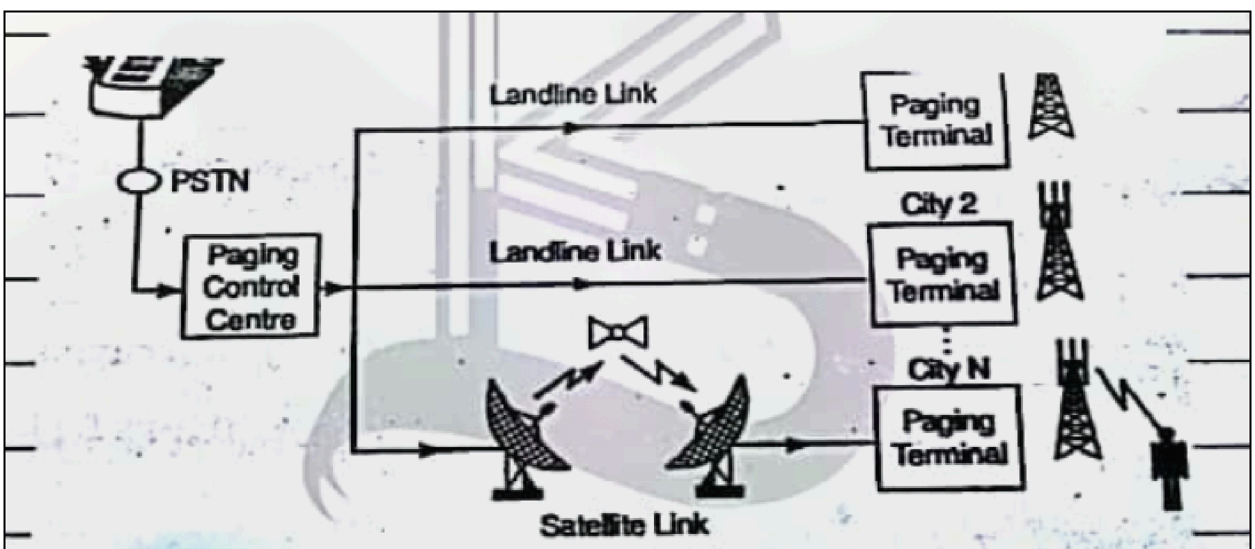
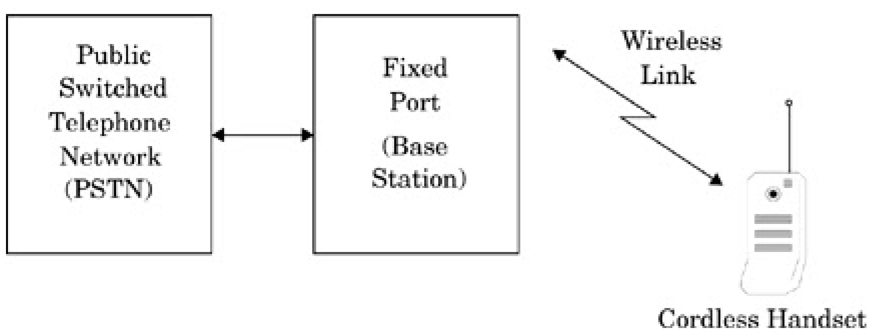


Fig. 1.5. A Wide Area Paging System.

Paging systems are one-way wireless messaging systems designed to provide reliable, wide-area communication to subscribers carrying a small receiver known as a pager.[1]

- **Functionality and Evolution:** Conventional paging systems were designed to send only brief, simple messages. Early pagers could only receive numeric messages, such as a phone number to call back. Modern paging systems evolved to support alphanumeric text, allowing for short text messages, news headlines, stock quotes, and even simple graphics to be delivered directly to the user's device.[1] The message sent to a subscriber is formally called a **page**.[1]
- **Simulcasting:** To ensure a page is received reliably across a large geographic area like a city, paging systems employ a technique called **simulcasting**. This involves a network of large, high-power radio towers that all broadcast the exact same page simultaneously. This massive redundancy ensures that the subscriber receives the message regardless of their location within the coverage area, overcoming issues like signal blockage from buildings.[1]
- **System Architecture:** A wide-area paging system is a centralized network. A call to a pager is first routed through the **Public Switched Telephone Network (PSTN)** to a central **Paging Control Centre**. This center processes the message and forwards it to **Paging Terminals** located in various cities. These terminals can be connected to the control center via dedicated landlines or satellite links. The terminals then transmit the page via the network of simulcasting towers to the subscriber's pager.[1]*A diagram illustrating the Wide Area Paging System Architecture should be included here. [FIG]*

Cordless Telephone Systems [FIG]



Cordless telephones, which first appeared in the 1970s, were designed to provide a low-cost, short-range wireless extension to a fixed landline telephone connection. They offer full-duplex voice communication between a portable handset and a dedicated base station.[1]

- **Purpose and Generations:**
  - **1st Generation (1G):** These early systems, popular in the 1980s, were intended for in-home use only. They provided a wireless link between the handset and its

    base unit over a very short range, typically just a few tens of meters.[1]
  - **2nd Generation (2G):** These systems were designed for both indoor and outdoor use, extending the range to a few hundred meters. They were often integrated with paging systems to receive messages and could be used in public spaces through shared base stations (telepoints).[1]
- **Digital Cordless Standards:** The move to digital technology brought several competing standards with advanced features. The development of these standards shows a parallel evolution to cellular systems, incorporating features like digital voice, power saving, and handoff, which were essential for any practical mobile device. This demonstrates that these were not just "cellular" problems but fundamental "mobile device" challenges.
  - **CT2 (Cordless Telephone, Second Generation):** Developed in Europe in 1989, CT2 was an early digital standard. It used FDMA with 40 channels and a Time Division Duplex (TDD) scheme for transmission. It supported a speech coding rate of 32 kbps.[1]
  - **DECT (Digital Enhanced Cordless Telecommunications):** A more advanced European standard that became globally popular. DECT uses a combination of TDMA and TDD (TDMA/TDD). It is notable for several advanced features:
    - **Sleep Mode:** To conserve battery power, the handset enters a low-power state when not in use.[1]
    - **Seamless Handoff:** DECT supports the ability to move a call from one base station to another without interruption, a crucial feature for mobility.[1]
    - **Time Slot Transfer:** To avoid interference, it can move an ongoing conversation from one time slot to another on the same channel.[1]
    - **GSM Compatibility:** DECT was designed to be compatible with the GSM cellular network, allowing for a degree of user mobility between the two systems.[1]
  - **PHS (Personal Handy Phone System):** A standard developed in Japan that offered telecommunication services for home, office, and outdoor environments. Like DECT, it used a TDMA/TDD scheme and included features like a sleep mode and handoff support.[1]
  - **PACS (Personal Access Communication System):** A standard developed in the United States, primarily designed for the "wireless local loop" (WLL) application, which is using wireless technology to replace the traditional copper wire connection to a home or office. It uses TDMA and uniquely supports both TDD and FDD modes of operation, as well as roaming management.[1]

A diagram showing the connection of a Cordless Handset to the PSTN via its Base Station should be included here. [FIG]

Tabular Comparison of Cordless Systems

The following table provides a detailed comparison of the key technical specifications for the major digital cordless telephone standards. This format allows for a quick analysis of their differences in terms of regional adoption, access methods, frequency bands, and performance characteristics.[1]

Parameter	CT-2 and CT-2(+)	DECT	PHS	PACS
Region	Europe/Canada	Europe	Japan	United States
Access Method	TDMA/TDD	TDMA/TDD	TDMA/TDD	TDMA/FDD
Frequency band (MHz)	864-868, 944-948	1,880-1,900	1,895-1,918	1,850-1,910, 1,930-1,990
Carrier spacing (kHz)	100	1,728	300	300
Bearer channels/carrier	1	12	4	8 per pair
Channel bit rate (kbps)	72	1,152	384	384
Modulation	GFSK	GFSK	π/4-DQPSK	π/4-DQPSK
Speech coding (kbps)	32	32	32	32
Average handset Tx power (mW)	5	10	10	25
Peak handset Tx power (mW)	10	250	80	200
Frame duration (ms)	2	10	5	2.5

The Cellular System Concept

The cellular concept is the foundational principle that enables modern mobile communication. By replacing a single, high-power transmitter with a network of low-power transmitters, it solves the critical problems of spectral inefficiency and limited capacity that plagued earlier mobile radio systems.

Core Principles: Cells, Clusters, and Frequency Reuse

The entire concept is built on three interconnected ideas: dividing the service area into cells, grouping these cells into clusters, and reusing frequencies across these clusters.[1]

- **Cell:** A cell is the smallest geographical unit of a cellular system. It represents the radio coverage area of a single Base Station (BS). Cells are typically depicted as hexagons in system diagrams because this shape tessellates a plane without gaps or overlaps, providing a convenient abstraction for network planning.[1]
- **Frequency Reuse:** This is the cornerstone of the cellular concept. A limited number of radio frequency channels are allocated to a service provider. Instead of using each channel only once across an entire city, the cellular system reuses the same set of channels in different cells that are geographically separated by a sufficient distance. This "reuse distance" is calculated to ensure that the interference between cells using the same frequencies (co-channel interference) remains below an acceptable level. This technique allows a network to support a vastly larger number of subscribers than would be possible with a conventional system.[1]
- **Cluster:** A cluster is a group of adjacent cells in which each cell is assigned a unique set of frequency channels. The total number of available channels in the system is divided among the cells within a single cluster. This entire cluster pattern, with its specific frequency allocation, is then repeated systematically across the entire service area to provide coverage. The number of cells in a cluster is known as the cluster size, denoted by N.[1]

A Basic Cellular System: Architecture and Components [FIG]

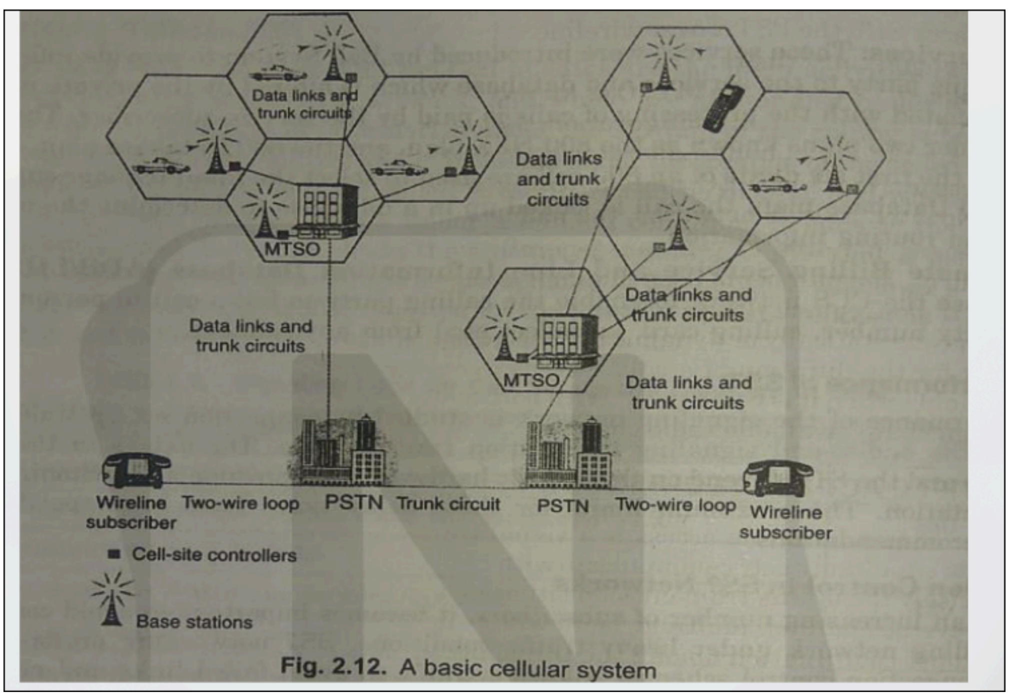


Fig. 2.12. A basic cellular system

A cellular system is composed of several key components that work together to provide a seamless connection from a mobile device to the global telephone network.[1]

- **Mobile Station (MS):** This is the user's terminal, such as a cellular phone or a data dongle. It contains a radio transceiver, an antenna, and processing circuitry to communicate with the Base Station over the air interface.[1]
- **Base Station (BS):** Often referred to as a cell site, the BS is a fixed station located within each cell. It houses the radio transmitters and receivers that handle all radio-related functions for the MSs in its cell. The BS acts as the bridge between the mobile user and the core network.[1]
- **Mobile Switching Center (MSC):** Also known as the Mobile Telephone Switching Office (MTSO), the MSC is the brain of the cellular network. It is a sophisticated telephone exchange that connects to a group of Base Stations. The MSC is responsible for:
  - **Call Processing:** Managing call setup, termination, and routing.
  - **Channel Assignment:** Allocating voice and control channels to the MS.
  - **Handoff Management:** Controlling the process of transferring a call from one cell to another as the user moves.
  - **PSTN Interconnection:** Providing the gateway connection to the Public Switched Telephone Network (PSTN), allowing mobile users to call landline phones and vice versa.[1]

A diagram illustrating the basic cellular system architecture, showing the interconnection between MS, BS, MSC, and PSTN, should be included here. [FIG]

The Air Interface: Channels

The **Common Air Interface (CAI)** defines the set of rules and logical channels for communication between the Mobile Station and the Base Station. These channels are divided based on their function and direction of transmission.[1]

- **Functional Division:**
  - **Control Channels:** These channels are used for signaling and system management. They carry information related to call initiation, channel assignment, handoff requests, and other control functions. They do not carry user voice or data.[1]
  - **Traffic Channels:** Also known as voice channels, these are used to carry the actual user conversation (voice) or data payload once a call has been established.[1]
- **Directional Division:**
  - **Forward Channels (Downlink):** These channels carry information from the Base Station to the Mobile Station. This includes the forward voice channel (FVC) and the forward control channel (FCC).[1]
  - **Reverse Channels (Uplink):** These channels carry information from the Mobile Station to the Base Station. This includes the reverse voice channel (RVC) and the reverse control channel (RCC).[1]

Advanced Cellular Concepts

To optimize capacity and coverage in real-world deployments, network planners use several advanced techniques that build upon the basic cellular model.[1]

- **Cell Splitting:** In urban areas or other locations with very high traffic density (e.g., stadiums, shopping malls), the capacity of a standard cell may be exceeded. Cell splitting is the process of dividing a large, congested cell into a group of smaller cells, each with its own low-power BS. This increases the number of times frequencies can be reused in a given area, thereby multiplying the overall capacity.[1]
- **Cell Sectoring:** Instead of using a single omnidirectional antenna at the BS to cover a cell in all directions, cell sectoring uses multiple directional antennas. Each antenna covers a specific sector of the cell (e.g., 120 degrees for a 3-sector cell). This technique reduces the amount of co-channel interference received from neighboring cells, which in turn allows for a smaller frequency reuse distance and thus higher system capacity.[1]
- **Umbrella Cell Pattern:** This is a hierarchical cell structure used to manage users moving at different speeds. A large "macrocell" (the umbrella) is overlaid on a group of smaller "microcells." Fast-moving users (e.g., in cars on a highway) are served by the large macrocell to minimize the number of handoffs required. Slow-moving or stationary users are served by the underlying microcells, which provide high capacity. The network intelligently assigns a user to the appropriate cell layer based on their speed.[1]

Generations of Cellular Systems

The history of cellular communication is marked by distinct "generations," each representing a major technological leap that introduced new capabilities, improved performance, and enabled new services. This section provides a comparative analysis of these generations.

First Generation (1G): Analog Systems

The first commercial cellular systems were analog and focused exclusively on providing voice communication.[1]

- **Technology:** 1G systems used analog frequency modulation (FM) for voice transmission. For multiple access, they employed **Frequency Division Multiple Access (FDMA)**, where the available spectrum was divided into distinct 30 kHz or 25 kHz channels, with one user per channel.[1]
- **Duplexing:** All 1G systems used **Frequency Division Duplex (FDD)**, allocating two separate frequency bands for the forward (BS to MS) and reverse (MS to BS) links to

enable simultaneous two-way conversation.[1]

- **Limitations:** Being analog, 1G systems had relatively low capacity, poor security (calls could be easily eavesdropped on with a radio scanner), and inconsistent voice quality.
- **Major Standards:**
  - **AMPS (Advanced Mobile Phone System):** Deployed in the Americas.
  - **ETACS (European Total Access Communication System):** Used in the United Kingdom and other parts of Europe.
  - **NTT (Nippon Telephone & Telegraph):** The standard used in Japan.[1]

Tabular Comparison of 1G Analog Systems

The following table compares the key parameters of the dominant 1G standards.[1]

Parameter	America (AMPS)	Europe (ETACS)	Japan (NTT)
Multiple Access	FDMA	FDMA	FDMA
Duplexing	FDD	FDD	FDD
Forward Channel (MHz)	869-894	935-960	870-885
Reverse Channel (MHz)	824-849	890-915	925-940
Channel Spacing (kHz)	30	25	25
Data Rate (kbps)	10	8	0.3
Spectral Efficiency (bps/Hz)	0.33	0.33	0.012
Capacity (Channels)	832	1000	600

Second Generation (2G): The Digital Revolution

The second generation marked a pivotal shift from analog to digital transmission, which brought significant improvements in capacity, security, and service offerings.[1]

- **Technology:** 2G systems introduced digital modulation and digital multiple access schemes, primarily **Time Division Multiple Access (TDMA)** and **Code Division Multiple Access (CDMA)**. This allowed multiple users to share a single frequency channel, dramatically increasing spectral efficiency.[1]
- **Advantages over 1G:**
  - **Increased Capacity:** Digital compression and multiple access techniques allowed for 3 to 10 times more users per channel than analog systems.
  - **Enhanced Security:** Digital signals could be easily encrypted, providing privacy

for conversations.[1]

  - **New Services:** The digital nature of 2G enabled the introduction of data services, most notably the **Short Message Service (SMS)**, or text messaging.[1]
  - **Improved Efficiency:** Lower transmit power requirements for mobile phones led to longer battery life and smaller devices.[1]
- **Major Standards:**
  - **GSM (Global System for Mobile Communication):** Originally a European standard, GSM (using TDMA) became the most widely adopted 2G standard in the world.[1]
  - **IS-54 (D-AMPS):** A TDMA-based standard used in North America, designed to be backward-compatible with the AMPS frequency plan.[1]
  - **IS-95 (cdmaOne):** A CDMA-based standard deployed in North America and parts of Asia.[1]
  - **PDC (Personal Digital Cellular):** A TDMA-based standard used exclusively in Japan.[1]

Tabular Comparison of 2G Digital Systems

This table contrasts the technical specifications of the major 2G standards, highlighting the different approaches to digital mobile communication.[1]

Parameter	US IS-54	Europe GSM	Japan PDC	US IS-95
Multiple Access	TDMA/FDD	TDMA/FDD	TDMA/FDD	CDMA/FDD
Modulation	π/4-DQPSK	GMSK	π/4-DQPSK	QPSK/BPSK
Forward Channel (MHz)	869-894	935-960	810-826	869-894
Reverse Channel (MHz)	824-849	890-915	940-956	824-849
Channel Spacing (kHz)	30	200	25	1250 (1.25 MHz)
Data/Chip Rate	48.6 kbps	270.833 kbps	42 kbps	1.2288 Mcps
Speech Codec Rate (kbps)	7.95	13	6.7	Variable (1.2-9.6)

Third Generation (3G): The Dawn of Mobile Data

The primary goal of 3G was to provide significantly higher data rates to enable a true mobile internet experience and support multimedia applications.[1]

- **Technology:** 3G systems are predominantly based on wideband **CDMA (W-CDMA)** technologies. They use wider channels (typically 5 MHz) to achieve higher data rates, with peak speeds of up to 2 Mbps.[1]
- **Services:** 3G enabled services such as mobile web browsing, video calling, streaming media, and location-based services.
- **Key Challenge - The Near-Far Problem:** A critical issue in CDMA systems is the "near-far problem." Because all users transmit on the same frequency, a mobile unit close to the base station can transmit at a power level that drowns out the much weaker signal from a user at the edge of the cell. To combat this, 3G systems rely on sophisticated and fast **power control** mechanisms, where the base station constantly instructs each mobile to adjust its transmit power so that all signals arrive at the receiver with roughly equal strength.[1]
- **Major Standards:**
  - **W-CDMA (UMTS):** The standard developed in Europe and widely adopted globally, representing an evolution of the GSM path.
  - **CDMA2000:** An evolution of the IS-95 standard, primarily used in North America and South Korea.
  - **TD-SCDMA:** A standard developed in China that uses Time Division Duplexing.[1]

Fourth Generation (4G) and Beyond

4G systems were designed to provide a leap forward in data speeds and an all-IP (Internet Protocol) network architecture, treating voice as just another data application.[1]

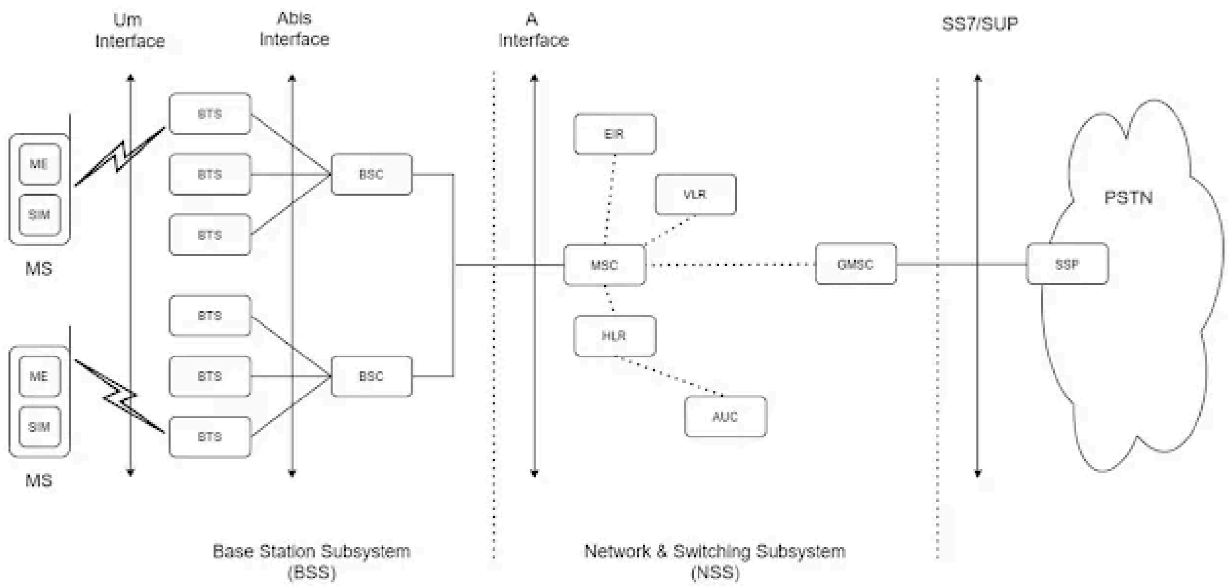
- **Goal:** The vision for 4G was defined by high usability ("anytime, anywhere, with any technology"), flexibility for the user to select services with a desired Quality of Service (QoS), and the seamless integration of various multimedia services at a low transmission cost.[1]
- **Core Concepts:** 4G focuses on providing very high data rates (tens to hundreds of Mbps), low latency, and global mobility support. It is characterized by personalization and a fully integrated, IP-based network architecture.[1]
- **Key Technologies:** While early definitions included enhancements of 3G technologies like WCDMA and CDMA2000, the primary air interface for true 4G systems (like LTE - Long-Term Evolution) is **Orthogonal Frequency-Division Multiple Access (OFDMA)**, which is highly efficient for high-speed data transmission in multipath environments.

Personal Communication Services (PCS) and Mobility Management

While cellular technology describes the underlying network structure, Personal

Communication Services (PCS) represents the user-centric vision of ubiquitous communication. Achieving this vision depends on sophisticated network functions that manage a user's location and movement seamlessly.

Personal Communication Services (PCS): Concept and Architecture [FIG]



PCS is a broad concept that refers to a wide variety of wireless services with the goal of enabling communication with any person, at any time, in any place, and in any form.[1]

- **Salient Features:**
  - **Global Roaming Capability:** The ability to use the service across different networks and countries.[1]
  - **Multimedia Services:** Support for high-quality voice, data, and video.[1]
  - **Single Personal Telecommunication Number (PTN):** A single number that can reach a person regardless of their location or the device they are using.[1]
  - **Universal Handset:** The idea of a single device that can work across multiple environments and networks.[1]
- **PCS Tiers:** PCS systems are often categorized into two tiers based on their coverage and mobility support:
  - **High-Tier PCS:** These are wide-area systems designed for high-speed vehicular and pedestrian mobility. This category includes standard cellular systems like GSM and CDMA.[1]
  - **Low-Tier PCS:** These are short-range, low-power systems designed for low-speed pedestrian or indoor environments. This category includes cordless

standards like DECT and PHS.[1]

- **Basic PCS Architecture:** A PCS network consists of two primary segments:
  1. **Radio Network:** This is the wireless part of the network that the user's device interacts with. It includes the **Mobile Station (MS)**, the **Base Station (BS)** that provides radio coverage in a cell, and the **Base Station Controller (BSC)** which manages a group of BSs.[1]
  2. **Wireline Transport Network:** This is the wired core network that connects the radio network to other networks. It consists of the **Mobile Switching Center (MSC)**, which performs call switching and control, and critical mobility databases.[1]
- **Mobility Databases:**
  - **Home Location Register (HLR):** This is a central, permanent database located in the user's home network. It stores the subscriber's master profile, including their service subscriptions, authentication information (like the International Mobile Subscriber Identity - IMSI), and their last known location (i.e., the address of the VLR currently serving them).[1]
  - **Visitor Location Register (VLR):** This is a temporary database located in the MSC of the network that a user is currently visiting (which could be their home network or a roaming partner's network). When a user enters a new MSC's area, the VLR retrieves a copy of their profile from their HLR. This allows the local MSC to handle call setup and other services without having to query the remote HLR for every transaction, making the network more efficient.[1]

A diagram illustrating the Basic PCS Network Architecture should be included here. [FIG]

Mobility Management: Handoff (Handover)

Handoff (or handover) is arguably the most critical function for enabling mobility. It is the process by which the network seamlessly transfers an active call or data session from one cell to another as the user moves, without any interruption to the service.[1]

- **Types of Handoff[FIG]:**
  - **Inter-BS Handoff (Intercell Handoff):** This is the most common type of handoff. It occurs when a mobile user moves from a cell served by one Base Station to an adjacent cell served by another Base Station, where both BSs are connected to the *same* MSC. The MSC orchestrates the entire process, instructing the MS to switch channels and re-routing the call path internally from the old BS to the new one.[1]
  - **Inter-System Handoff (Inter-MSC Handoff):** This is a more complex handoff that occurs when a user moves across the boundary of two service areas controlled by *different* MSCs. This process requires significant signaling and coordination between the original MSC and the new MSC to transfer control of the call and establish a new connection path through the new MSC.[1]

    - **Path Minimization:** In scenarios involving multiple inter-system handoffs (e.g., a user moves from an area controlled by MSC-A to MSC-B, and then to MSC-C), the call path can become inefficiently long (A -> B -> C). Path minimization is a procedure where the network intelligently re-routes the connection to be more direct (A -> C), releasing the resources on the intermediate MSC-B and improving network efficiency.[1]

Diagrams illustrating the Inter-BS Handoff process, the Inter-System Handoff process, and Path Minimization should be included here. [FIG]

Handoff Detection and Execution

The decision of when and how to perform a handoff is crucial for network performance. This process involves detecting the need for a handoff, selecting a target cell, and managing channel resources.

- **Handoff Detection Strategies:** The evolution from network-controlled to mobile-assisted handoff reflects a broader trend of decentralizing network intelligence to the edge devices. Early systems with large cells could afford the delay of a centralized decision process (NCHO). However, modern high-density networks with small microcells require the speed and efficiency of a distributed process where the mobile device actively participates (MAHO), making it a critical enabler of today's network performance.
    - **Network Controlled Handoff (NCHO):** In this early strategy, the network is solely responsible for the handoff decision. The BSs monitor the signal strength of the mobile unit. When the signal drops below a certain threshold, the MSC polls surrounding BSs to find a better candidate and then directs the handoff. This process is slow (can take up to 10 seconds) and creates a heavy signaling load on the network. It was used in 1G systems like AMPS.[1]
    - **Mobile Controlled Handoff (MCHO):** In this strategy, the mobile station takes full control. The MS continuously monitors the signal from its current BS and neighboring BSs. When handoff criteria are met, the MS itself initiates the handoff to the best candidate cell. This is common in low-tier, short-range systems like DECT and PACS.[1]
    - **Mobile Assisted Handoff (MAHO):** This is a hybrid and the most common modern approach. The network still makes the final decision, but it is assisted by the mobile station. The MS constantly measures the signal strength of nearby cells and reports these measurements back to its serving BS. The network uses this real-time data to make a fast and accurate handoff decision. MAHO is significantly faster (1-2 seconds) and more efficient than NCHO and is used in all modern systems like GSM and CDMA/W-CDMA.[1]
- **Channel Assignment Strategies:** When a new call is initiated or a handoff occurs, the network must assign a voice channel.

- **Fixed Channel Assignment (FCA):** Each cell is allocated a predetermined, fixed set of voice channels. If a new call arrives and all channels in that cell are occupied, the call is blocked. This strategy is simple to implement but is inefficient, as it cannot adapt to changing traffic patterns.[1]
- **Borrowing Channel Assignment (BCA):** A variation of FCA where a cell that has used all its channels can borrow a free channel from a neighboring cell, provided it does not cause interference. This process is supervised by the MSC and adds some flexibility.[1]
- **Dynamic Channel Assignment (DCA):** In this strategy, channels are not permanently allocated to cells. Instead, they are kept in a central pool managed by the MSC. When a channel is needed, the MSC dynamically allocates one to the cell for the duration of the call. This strategy is highly efficient and reduces call blocking, but it significantly increases the computational and signaling load on the MSC.[1]

- **Handoff Prioritization Schemes:** From a user's perspective, having an ongoing call forcibly terminated (a dropped call due to handoff failure) is far more disruptive than having a new call attempt blocked. Therefore, networks implement strategies to give handoff requests priority over new call requests.
  - **Reserved Channel Scheme (RCS) / Guard Channels:** This is the most common method. In each cell, a small number of channels are reserved exclusively for handoff calls. New calls are only granted access to the non-reserved channels. This significantly reduces the probability of handoff failure, at the cost of slightly increasing the blocking probability for new calls.[1]
  - **Queuing Prioritizing Scheme:** This scheme takes advantage of the fact that there is usually an overlap region between cells where a mobile can communicate with either base station for a short period. If a handoff request arrives and no channels are free, instead of being dropped immediately, the request is placed in a queue. The network will attempt to serve the queued handoff request as soon as a channel becomes free, giving it priority over any new call attempts.[1]

Roaming Management

Roaming is the facility that allows a mobile subscriber to automatically access services (make and receive calls, use data) when they are traveling outside the geographical coverage area of their home network, by using the network of another service provider.[1]

- **Registration (Location Update)[FIG]:** This is the fundamental process that enables roaming. When a mobile device is turned on in a foreign network (a "visited network"), it performs a location update. The steps are as follows:
  1. The MS registers with the VLR of the visited network's MSC.
  2. The new VLR contacts the user's HLR in their home network to inform it of the user's current location and to request the user's subscription profile.

3. The HLR authenticates the user and sends a copy of their profile to the new VLR. It also updates its own records to point to this new VLR.
4. The HLR sends a de-registration message to the user's old VLR (where they were previously located) to cancel the old location record.
5. Once the new VLR has the user's profile, it informs the MS that registration was successful, and the user can now use the visited network.[1]

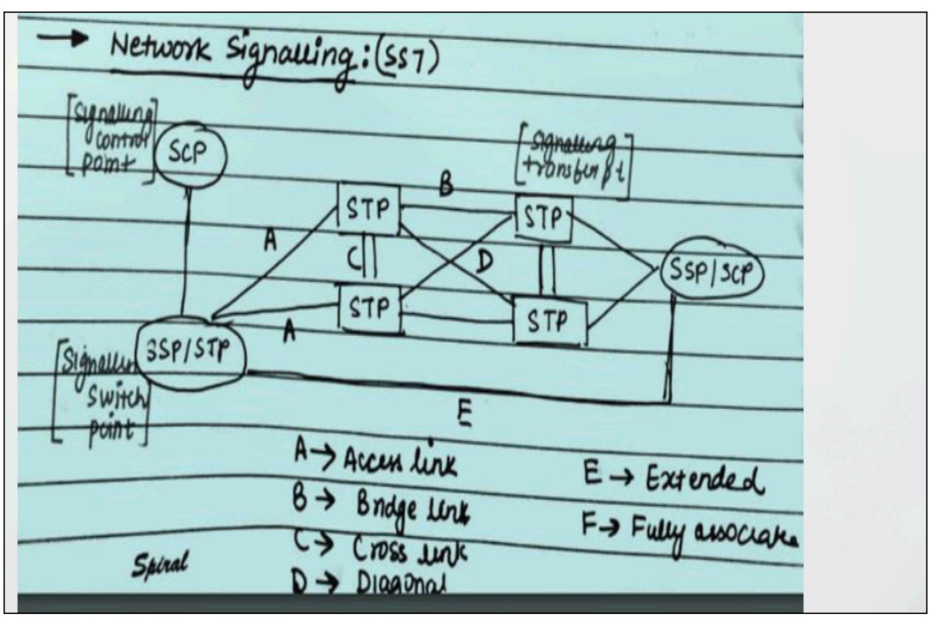
A diagram illustrating the Roaming Management and Location Update process should be included here. [FIG]

- **Location Tracking (for Call Delivery):** This is the process the network uses to find a roaming user to deliver an incoming call.
  1. A call is placed to the mobile user's number. The call is routed to their home network's MSC (known as the Gateway MSC).
  2. The Gateway MSC queries the user's HLR to find their current location.
  3. The HLR provides the address of the VLR/MSC in the visited network where the user is currently registered.
  4. The Gateway MSC then routes the call to the visited MSC, which in turn pages the mobile device in its current cell to complete the call.[1]

Network Signaling and Multiple Access

This section explores the underlying mechanisms that control network operations and enable the sharing of the radio spectrum. Network signaling protocols are the "language" that network components use to communicate, while multiple access techniques are the fundamental rules for sharing the airwaves.

Network Signaling: The SS7 Protocol[FIG]



Signaling System No. 7 (SS7) is a set of international telecommunications protocols used to set up and tear down most of the world's public telephone calls. In cellular networks, it is the backbone for communication between MSCs and other network elements like the HLR and VLR, and is essential for enabling roaming and advanced call features.[1]

- **SS7 Architecture Components:** The SS7 network is a packet-switched network that runs parallel to the voice network. Its main components are:
  - **Signaling Switching Point (SSP):** These are telephone switches (like an MSC) equipped with SS7 software. They originate, terminate, or switch signaling messages.[1]
  - **Signaling Transfer Point (STP):** These are the packet routers of the SS7 network. They route signaling messages between SSPs and SCPs based on their destination address.[1]
  - **Signaling Control Point (SCP):** These are databases that provide access to advanced network services. In a mobile context, the HLR and VLR are examples of SCPs. An MSC (acting as an SSP) will send a query to an SCP to get information needed to process a call (e.g., "Where is this roaming user?").[1]

  A diagram of the SS7 Network Architecture, showing the interconnection of SSP, STP, and SCP nodes, should be included here. [FIG]
- **SS7 Protocol Architecture:** The SS7 protocol is a layered stack, similar in concept to the OSI model.

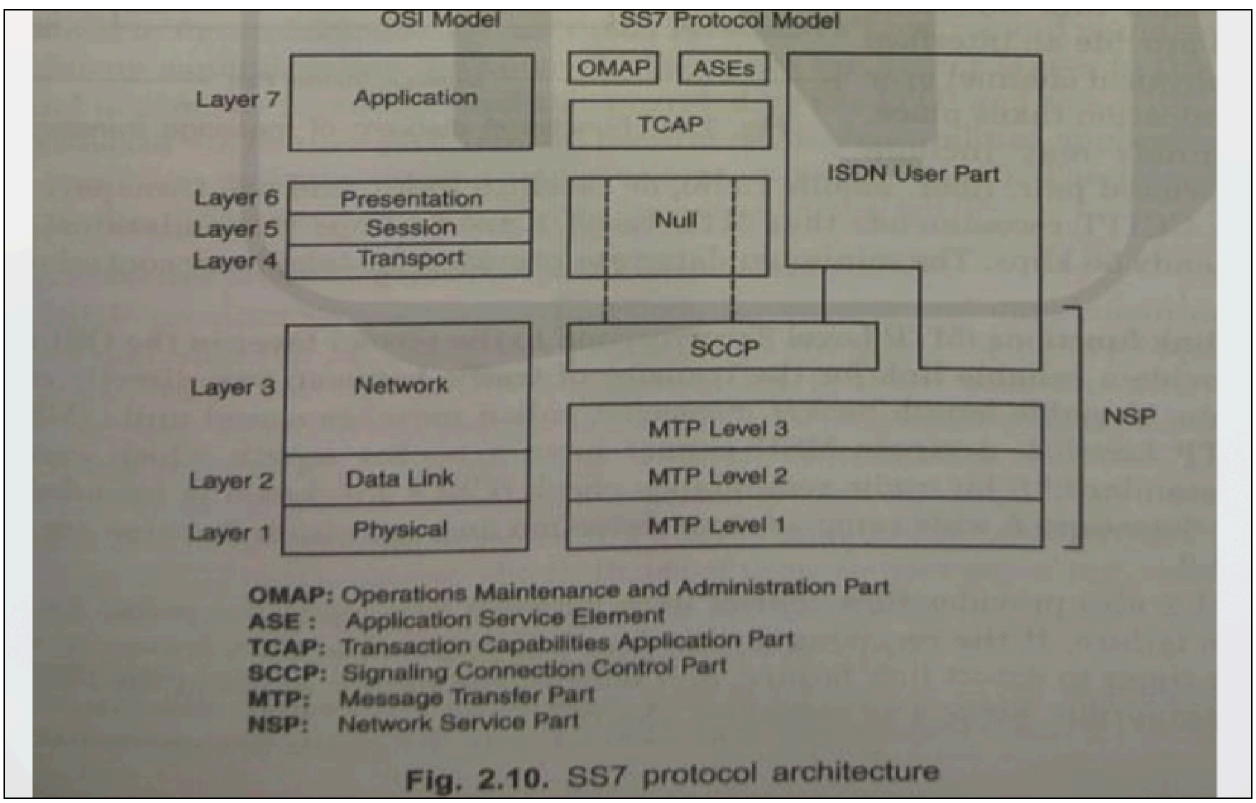


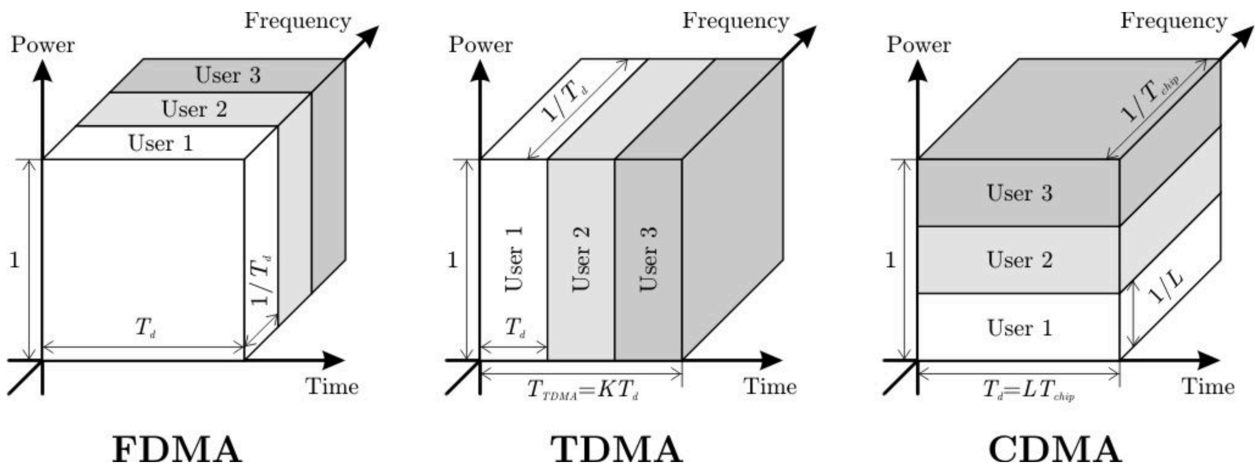
Fig. 2.10. SS7 protocol architecture

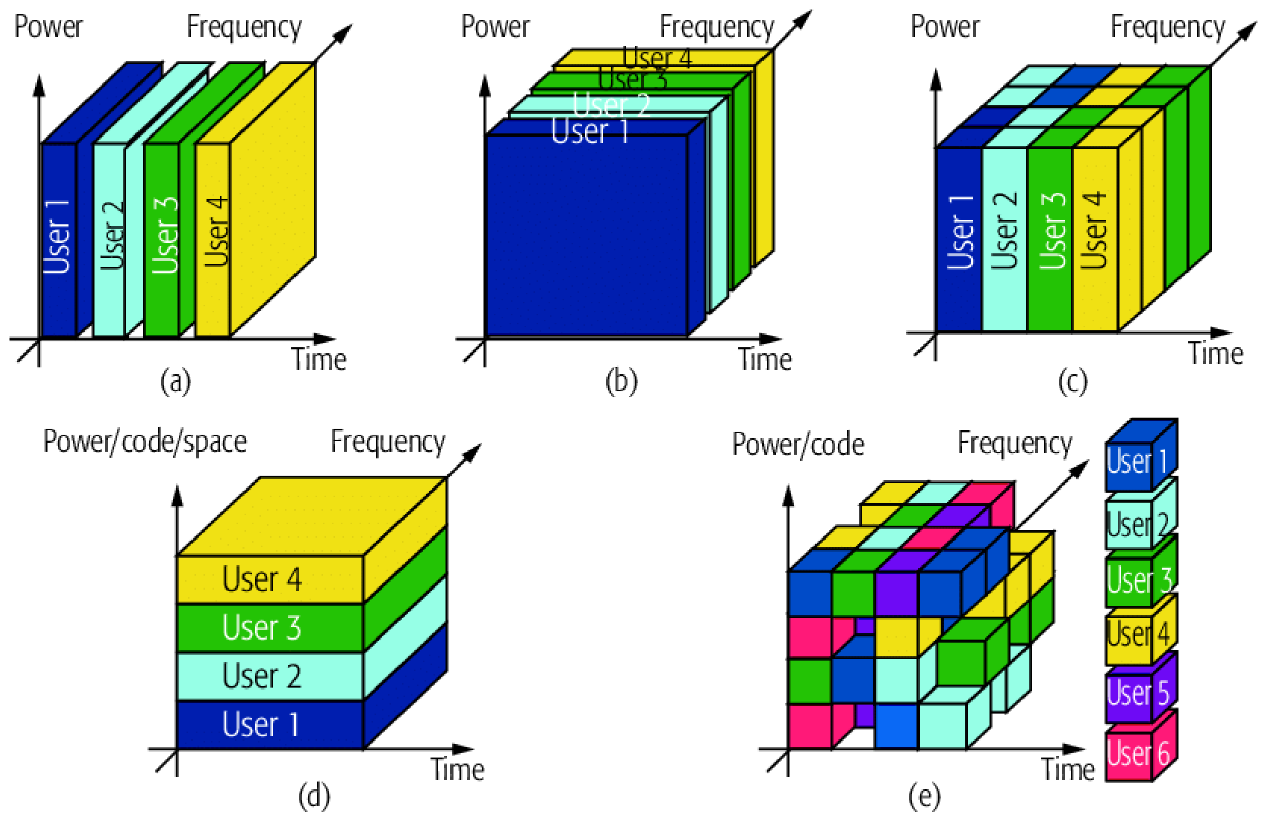
  - **Message Transfer Part (MTP):** MTP Levels 1, 2, and 3 form the foundation of the protocol. They are responsible for the reliable, node-to-node transport of signaling messages across the network. They correspond roughly to the Physical, Data Link, and Network layers of the OSI model.[1]

  - **Signaling Connection Control Part (SCCP):** This layer sits on top of MTP-3 and provides enhanced addressing capabilities, allowing messages to be routed to specific applications within a signaling node.[1]
  - **User Parts:** These are the application layer protocols that use the transport services of MTP and SCCP. The key user parts in mobile networking are:
    - **ISDN User Part (ISUP):** Used for setting up and tearing down basic voice and data calls between switches.[1]
    - **Transaction Capabilities Application Part (TCAP):** Used for non-circuit-related information exchange, typically for database queries. For example, when an MSC needs to query an HLR for a subscriber's location, it uses a TCAP message.[1]

A diagram comparing the SS7 Protocol Stack to the OSI Model should be included here. [FIG]

Multiple Access Techniques





Multiple access schemes are fundamental to cellular communication, as they define how a limited amount of radio spectrum can be shared among many users simultaneously.[1]

- **Frequency Division Multiple Access (FDMA):** In FDMA, the available frequency spectrum is divided into a number of narrower frequency channels. Each user is allocated an exclusive channel for the duration of their call. To prevent interference between adjacent channels, small unused frequency strips called **guard bands** are placed between them. FDMA is conceptually simple but can be inefficient if a user's channel is idle.[1]
- **Time Division Multiple Access (TDMA):** In TDMA, multiple users share the same, wider frequency channel, but they take turns using it. The time on the channel is divided into frames, and each frame is further divided into a number of time slots. Each user is assigned a specific time slot within the repeating frame. The transmission is therefore not continuous, but occurs in bursts. This requires precise synchronization but is more spectrally efficient than FDMA for digital traffic.[1]
- **Code Division Multiple Access (CDMA):** In CDMA, all users transmit on the same frequency at the same time. Users are separated not by frequency or time, but by a unique mathematical code (a pseudo-random sequence). The transmitter spreads the user's signal across a wide frequency band using this code. The receiver, knowing the code, can then "de-spread" the signal and recover the original data, while the signals from all other users (using different codes) appear as background noise. CDMA offers high capacity and features like soft handoff, but it requires very strict power control to mitigate the **near-far problem**.[1]

- **Space Division Multiple Access (SDMA):** SDMA is a more advanced technique that reuses spectrum based on spatial separation. It uses "smart antennas" at the base station that can form highly directional beams of radio energy, pointing a specific beam directly at each user. By creating these narrow "spot beams," the base station can communicate with multiple users on the same frequency and at the same time, as long as they are in different locations (different "spaces"). SDMA is typically used in conjunction with FDMA, TDMA, or CDMA to further enhance system capacity.[1]

Tabular Comparison of Multiple Access Techniques

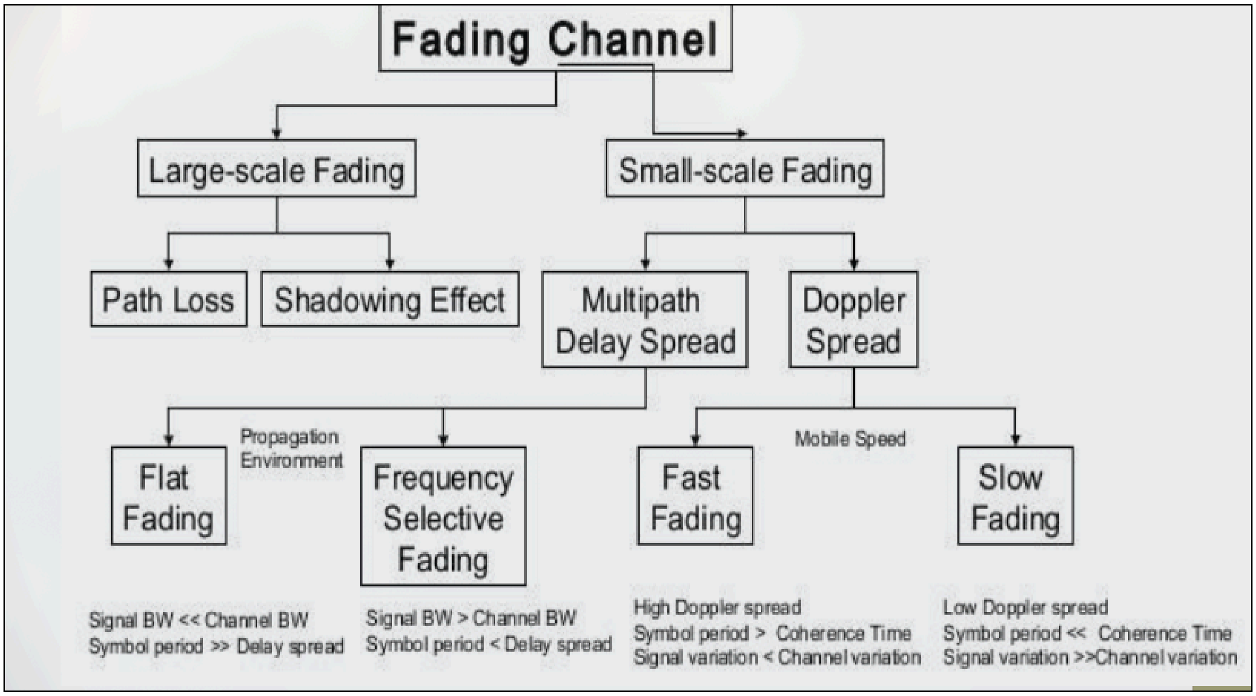
This table provides a comprehensive comparison of the four primary multiple access schemes, highlighting their core principles, advantages, and disadvantages.[1]

Parameter	FDMA	TDMA	CDMA	SDMA
Core Idea	Segment frequency band into disjoint sub-bands.	Segment sending time into disjoint time slots.	Spread the signal across the full spectrum using orthogonal codes.	Segment space into cells/sectors using directed antennas.
Signal Separation	Filtering in the frequency domain.	Synchronization in the time domain.	Code correlation at the receiver.	Cell structure and directed antenna beams.
Transmission	Continuous	Discontinuous (Bursty)	Continuous	Continuous
Handoff Type	Hard	Hard	Soft	Potentially Soft
User Terminal Complexity	Simple	Medium	More Complex	Requires smart antennas (at BS)
Key Advantage	Simple, robust, well-established.	Fully digital, flexible data rates.	High capacity, less frequency planning, soft handoff.	Very simple concept, increases capacity per km^2.
Key Disadvantage	Inflexible, frequencies are a scarce resource.	Requires strict synchronization and guard times.	Complex receivers, requires strict power control (near-far problem).	Inflexible if antennas are fixed; only useful with other access methods.

The Wireless Channel and Diversity

The physical medium through which radio waves travel is not perfect. The wireless channel is a harsh and unpredictable environment that distorts the signal in various ways. This section examines the nature of these impairments and the sophisticated techniques, known as diversity, that are used to overcome them and ensure reliable communication.

Wireless Propagation and Fading



Unlike a wired channel (like a fiber optic cable) which is stable and predictable, the wireless channel is constantly changing due to the environment and user mobility.[1]

- **Multipath Propagation:** A transmitted radio signal rarely travels in a straight line to the receiver. Instead, it propagates along multiple paths simultaneously due to three main mechanisms:
  - **Reflection:** The signal bounces off large, smooth surfaces relative to its wavelength, such as the ground, buildings, or walls.[1]
  - **Diffraction:** The signal bends around sharp edges of obstacles, such as the corners of buildings or the tops of hills. This allows the signal to reach a receiver even when there is no direct line-of-sight (LOS) path.[1]
  - **Scattering:** The signal strikes objects that are small relative to its wavelength, such as foliage, street signs, or lamp posts, causing the energy to be dispersed in many directions.[1]

    The result is that the receiver sees a sum of multiple copies of the original signal, each arriving at a slightly different time, with a different amplitude and phase.1
- **Fading:** The constructive and destructive interference of these multipath components causes the received signal strength to fluctuate rapidly. When the components add up

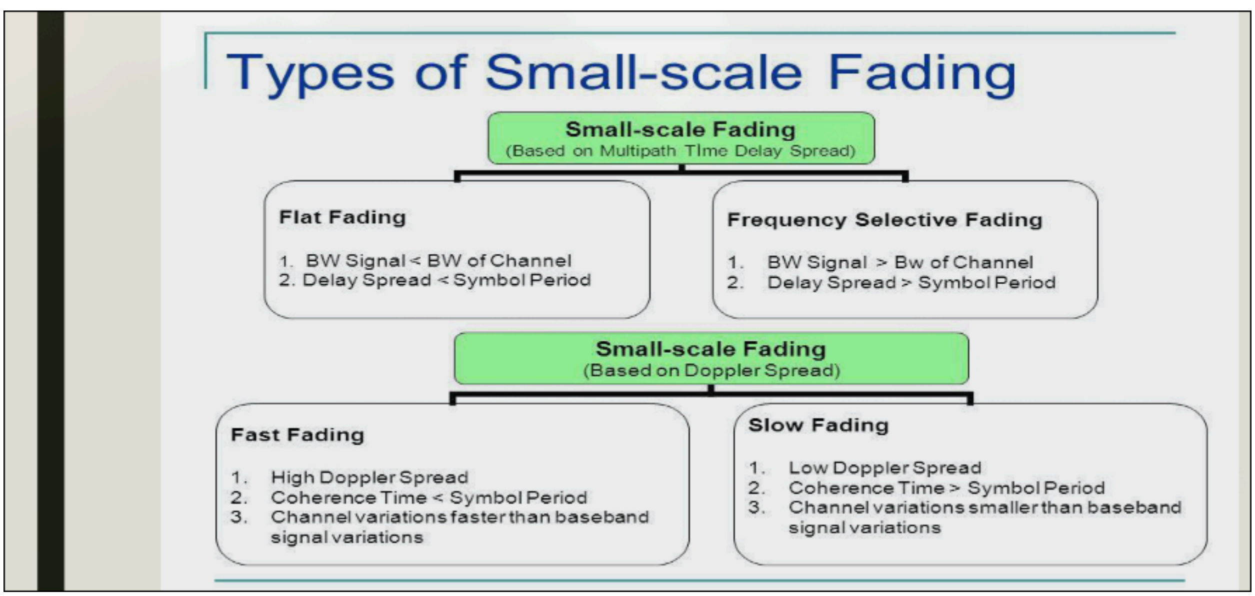
in phase, the signal strength increases; when they add up out of phase, they can cancel each other out, causing a deep drop in signal strength known as a **fade**.[1]

- **Large-Scale Fading:** This refers to the gradual variation of the average signal power over long distances (hundreds or thousands of meters). It is caused by two effects:
  - **Path Loss:** The natural attenuation of signal strength as it propagates over distance.
  - **Shadowing:** Signal blockage by large obstacles like buildings or hills, causing significant drops in average power over large areas.[1]
- **Small-Scale Fading:** This refers to the very rapid fluctuations in signal amplitude and phase over short distances (on the order of a half-wavelength) or short time durations. It is caused directly by the local multipath interference and is often referred to as **Rayleigh Fading**.[1]
- Doppler Shift: When the receiver or transmitter (or objects in the environment) are in motion, it causes a shift in the frequency of the received signal. This Doppler shift, fd, is given by the formula:

  $fd=\lambda v\cos\theta$

  where v is the velocity, λ is the signal's wavelength, and θ is the angle between the direction of motion and the direction of signal arrival. This effect is a primary cause of rapid channel variations.1

Types of Small-Scale Fading

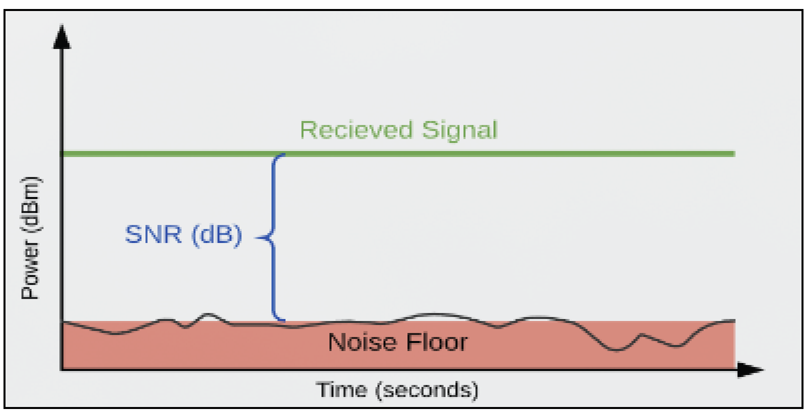


Small-scale fading is classified based on how the channel's characteristics relate to the characteristics of the transmitted signal.[1]

- **Based on Multipath Time Delay Spread:**

- **Flat Fading:** This occurs when the bandwidth of the transmitted signal (Bs) is much *less than* the coherence bandwidth of the channel (Bc). In this case, all frequency components of the signal experience the same amount of fading simultaneously. The channel acts like a flat attenuator across the signal's spectrum. This is typical for narrowband signals.[1]
- **Frequency Selective Fading:** This occurs when the signal's bandwidth is *greater than* the channel's coherence bandwidth (Bs>Bc). Different frequency components of the signal will experience different amounts of fading. This distorts the received signal and can cause **Inter-Symbol Interference (ISI)**, where the delayed copies of one symbol interfere with the reception of subsequent symbols. This is typical for wideband signals.[1]

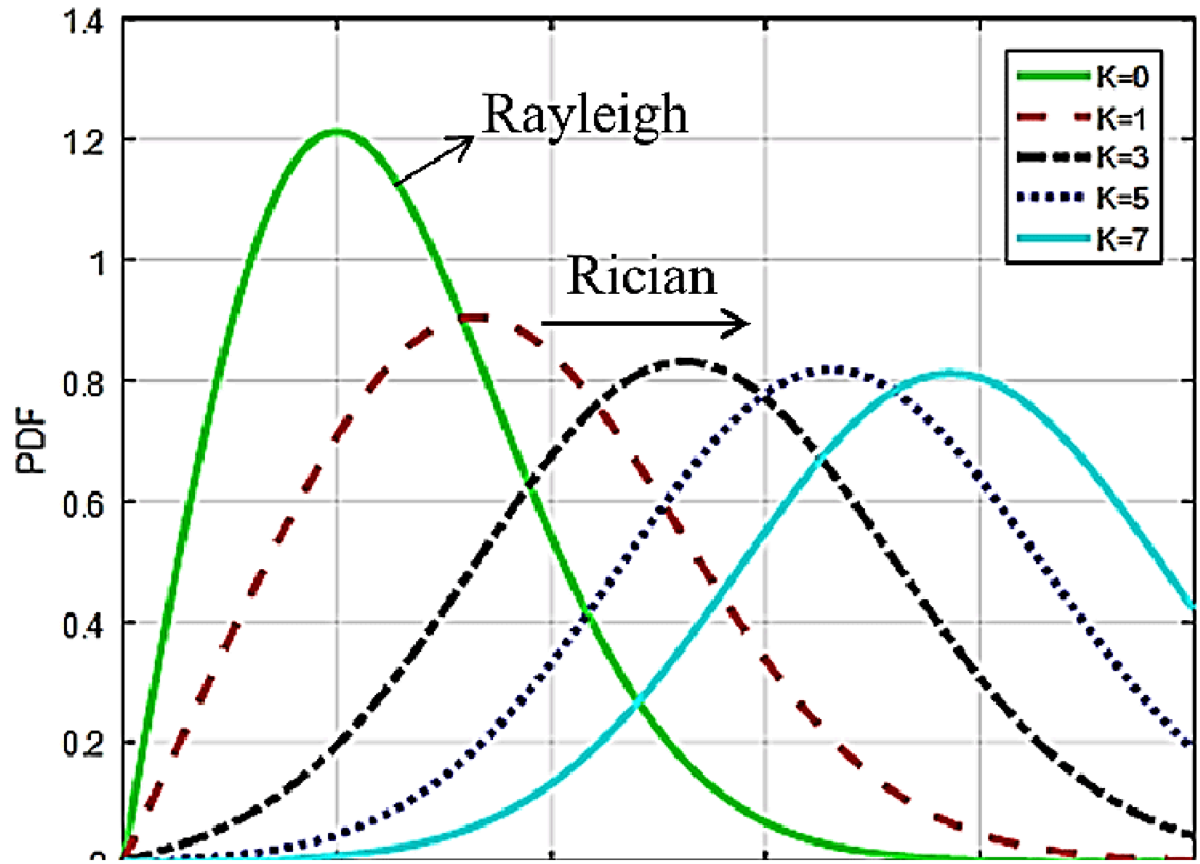
- **Based on Doppler Spread:**
    - **Fast Fading:** This occurs when the channel's impulse response changes rapidly *within* the symbol duration of the signal. This happens when the coherence time of the channel (Tc) is *less than* the symbol period (Ts). This is caused by high Doppler spread (i.e., high user velocity). Fast fading can severely distort the signal shape.[1]
    - **Slow Fading:** This occurs when the channel changes much more slowly than the transmitted signal. The coherence time of the channel is *greater than* the symbol period (Tc>Ts). The channel can be considered static for the duration of one or more symbols. This is caused by low Doppler spread (low user velocity).[1]



Statistical Fading Models and Performance

To analyze the effects of fading, statistical models are used to describe the random nature of the received signal's envelope.

- **Rayleigh Fading Model:** This is the most common model for small-scale fading. It is used in environments where there is **no dominant Line-of-Sight (NLOS)** path between the transmitter and receiver. The received signal is the sum of a large number of independent scattered components, and its envelope follows a Rayleigh probability distribution. This model represents a "worst-case" fading scenario.[1]



❑ The Rayleigh pdf is

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & for\ r > 0 \\ 0 & otherwise \end{cases}$$

Where r is the envelope amplitude of Rx signal & $2\sigma^2$ is the power of the signal

- **Rician Fading Model:** This model is used when there is a **dominant, strong Line-of-Sight (LOS)** path present, in addition to the scattered multipath components. The presence of this stable LOS component makes the fading less severe than in the Rayleigh case. The envelope of the received signal follows a Rician distribution. As the strength of the LOS component increases relative to the scattered components, the Rician channel approaches a simple non-fading Additive White Gaussian Noise (AWGN) channel.[1]

The fading is referred to as *Rician fading*

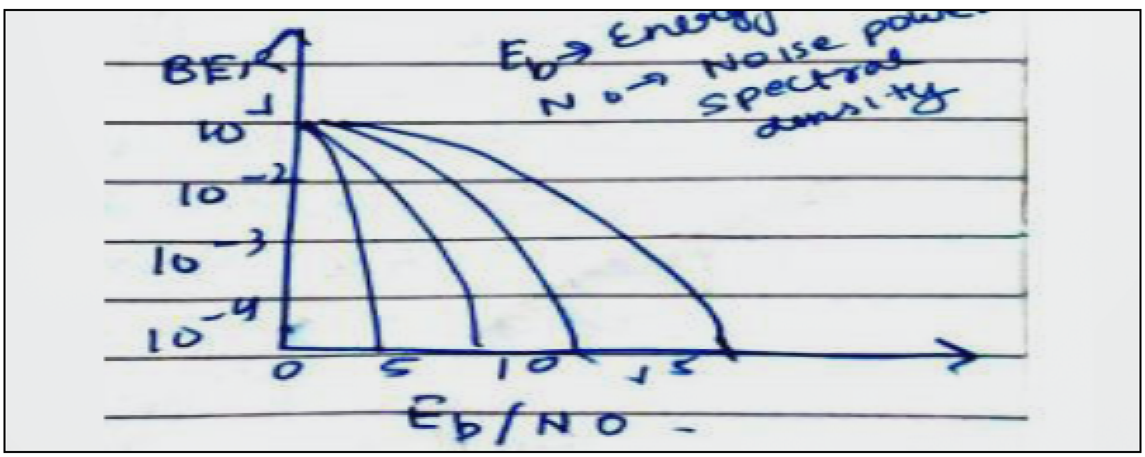
$$p(r_0) = \begin{cases} \dfrac{r_0}{\sigma^2} \exp\left[-\dfrac{(r_0^2 + A^2)}{2\sigma^2} \right] I_0\left(\dfrac{r_0 A}{\sigma^2}\right) & \text{for } r_0 \geq 0,\ A \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

A: the peak amplitude of the non fading signal component called (*specular component*)

σ^2: the predetection mean power of the multipath signal.

I_0 : the modified Bessel function of the first kind and zero order.

- **BER Performance in Fading Channels [FIG]:** The performance of a digital communication system is measured by its Bit Error Rate (BER). A plot of BER versus the Signal-to-Noise ratio per bit (Eb/No) reveals the severe impact of fading.
  - For a given target BER, a fading channel requires a much higher Eb/No than a simple AWGN (non-fading) channel.
  - The **AWGN channel** provides the best performance (lowest BER for a given Eb/No).
  - The **Rician channel** performs worse than AWGN but significantly better than Rayleigh, due to the stabilizing effect of the LOS path.
  - The **Rayleigh channel** provides the poorest performance, requiring a very high signal-to-noise ratio to achieve even a modest BER.[1]



A graph showing the comparative BER vs. Eb/No performance curves for AWGN, Rician, and Rayleigh fading channels should be included here. [FIG]

Diversity Techniques for Fading Mitigation

To combat the detrimental effects of fading, especially deep fades that can cause a total loss of signal, wireless systems employ **diversity**.

- **Principle of Diversity:** The core idea of diversity is to provide the receiver with multiple, independently faded replicas of the same transmitted signal. The rationale is that the probability of all replicas being in a deep fade simultaneously is much lower than the probability of a single signal being in a fade. By intelligently combining these replicas, the receiver can create a more robust and reliable composite signal.[1] *A diagram illustrating the Principle of Diversity with multiple independent links between transmitter and receiver should be included here. [FIG]*

- **Microscopic vs. Macroscopic Diversity:**
  - **Microscopic Diversity:** This is used to combat small-scale, fast fading. It involves creating independent signal paths over small distances or time scales, for example, by using multiple antennas separated by a few wavelengths.[1]
  - **Macroscopic Diversity:** This is used to combat large-scale, shadow fading. It involves selecting a signal path that is not physically obstructed. The most common example is a handoff, where the system switches to a different base station that is not shadowed from the user's perspective.[1]
- **Diversity Methods:**
  - **Space Diversity (or Antenna Diversity):** This is one of the most common and effective methods. It uses multiple antennas that are spatially separated at the receiver (or transmitter). As long as the antennas are separated by at least half a wavelength, the signals they receive will experience largely independent fading, providing diversity.[1]

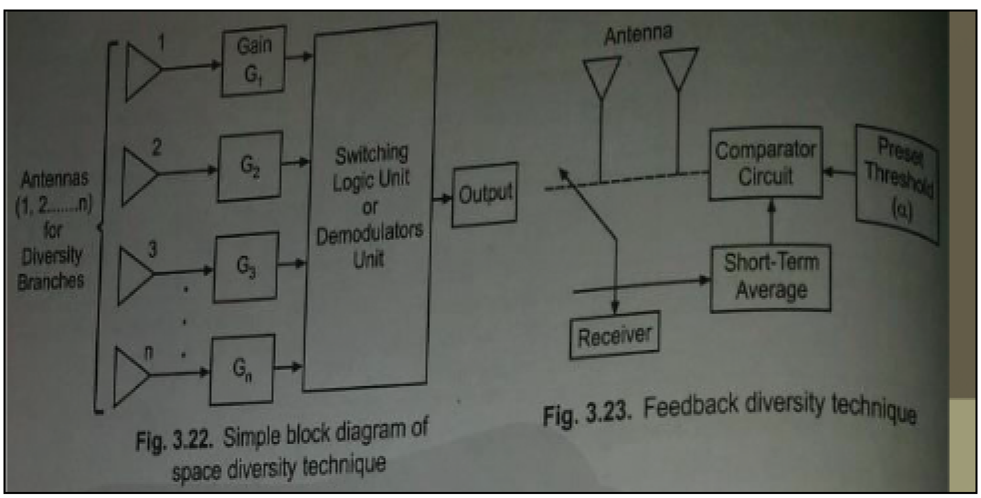


Fig. 3.22. Simple block diagram of space diversity technique

Fig. 3.23. Feedback diversity technique

  - **Frequency Diversity:** The same information is transmitted simultaneously on multiple carrier frequencies. The frequencies must be separated by more than the channel's coherence bandwidth to ensure they fade independently. This method is effective but inefficient as it consumes extra spectrum.[1]
  - **Time Diversity:** The same information is transmitted at different points in time. The time separation must be greater than the channel's coherence time. This is often achieved through channel coding and interleaving. It is effective but introduces latency.[1]
  - **Polarization Diversity:** This technique uses two antennas with orthogonal polarizations (e.g., one vertically polarized and one horizontally polarized).

    Reflections and scattering in the channel can alter the polarization of the signal, so the two antennas receive independently faded components. It is useful at base stations where space for antenna separation may be limited.[1]

  Diagrams showing a block diagram of a Space Diversity system and a Polarization Diversity antenna setup should be included here. [FIG]

- **Space Diversity Combining Techniques:** Once the receiver has multiple signal copies from its space diversity antennas, it must combine them.
  - **Selection Combining (SC):** This is the simplest technique. At any given moment, the receiver circuitry measures the Signal-to-Noise Ratio (SNR) of each branch and simply selects the one branch with the highest SNR to feed to the demodulator. The other branches are ignored.[1]
  - **Maximal Ratio Combining (MRC):** This is the optimal combining technique. The signals from all diversity branches are co-phased (their phase shifts are aligned) and then weighted individually according to their SNR (stronger signals get higher weights). Finally, they are summed together. This method maximizes the SNR of the combined signal, providing the best performance but also having the highest complexity.[1]
  - **Equal Gain Combining (EGC):** This is a compromise between SC and MRC. The signals from all branches are co-phased and then summed together with equal weights (i.e., a weight of 1). It is less complex than MRC because it does not require estimating the SNR of each branch, but its performance is only slightly worse than MRC and significantly better than SC.[1]
- **The Rake Receiver:** The Rake receiver is a specialized receiver architecture designed specifically for **CDMA** systems operating in a multipath environment. It brilliantly turns the problem of multipath into an advantage by using it as a form of diversity.
  - **Functionality:** A Rake receiver has multiple "fingers." Each finger is a separate correlator that can be synchronized to a different multipath component of the received signal. It essentially "rakes in" the signal energy from the multiple delayed copies. By treating each multipath component as an independent diversity branch, the receiver can combine them (typically using MRC) to create a much stronger and more reliable signal. This makes CDMA systems highly robust against multipath fading.[1]